11.3 Metis AIPU: A 12nm 15TOPS/W 209.6TOPS SoC for Cost- and Energy-Efficient Inference at the Edge

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The Metis AI Processing Unit (AIPU) is a quad-core System-on-Chip (SoC) designed for edge inference, executing all components of an AI workload on-chip. The Metis AIPU exhibits performance of 52.4 TOPS per AI core, and a compound throughput of 209.6 TOPS. Key features of the Metis AIPU and its integration into a PCIe card-based system are shown in Fig. 11.3.1. Metis leverages the benefits from a quantized digital in-memory computing (D-IMC) architecture — with 8b weights, 8b activations, and full-precision accumulation — to decrease both the memory cost of weights and activations and the energy consumption of matrix-vector multiplications (MVM), without compromising the neural network accuracy.

The Metis AIPU top-level architecture, illustrated in Fig. 11.3.2, incorporates four AI cores integrated into a SoC. This SoC comprises a RISC-V system controller, a security module, PCIe Gen 3, LPDDR4x, and a sizeable on-chip SRAM. All is interconnected via a high-speed, packetized Network-on-Chip (NoC). The RISC-V controller boots the chip, interfaces with peripherals, and manages the AI cores using a real-time OS. The PCIe offers a high-speed connection to an external host for offloading neural network tasks to the Metis AIPU. The NoC links the AI cores to a shared memory system that consists of a 32 MiB L2 SRAM and several GiB of an optional LPDDR4x SDRAM. Including the 4 MiB L1 and the 1 MiB D-IMC SRAM of each of the four AI-Cores, the on-chip SRAM aggregates to 52 MiB. The NoC splits control and data transfers, tailored to reduce contention for simultaneous accesses of multiple data managers. It offers more than a terabit per second of aggregated bandwidth to the shared memories, ensuring the AI cores will not stall in highly congested multi-core scenarios.

The primary component of the Metis AIPU is the AI Core, comprising the D-IMC for MVM operations, a data processing unit (DPU) for element-wise vector operations and activations, a depth-wise processing unit (DWPU) for depth-wise convolution, pooling, and up-sampling, a local 4 MiB L1 SRAM, and a RISC-V control core. The block diagram of the AI Core and its operational model are illustrated in Fig. 11.3.3. This core is a RISC-V-controlled dataflow engine featuring dual high-throughput streaming data paths and ensuring balanced performance across the diverse nature of modern neural network workloads. The AI Core is designed to execute all the layers of a neural network independently, eliminating the need for external interactions. The AI-Cores provide flexibility in deployment: they can jointly tackle a workload to enhance throughput, work simultaneously on the same neural network to cut down latency, or manage different neural networks independently in multi-network applications. Within each AI core is a large-scale D-IMC based MVM engine, designed to accelerate matrix operations. In-memory computing (IMC) is an emerging paradigm wherein crossbar arrays of memory devices are used to store a matrix and perform MVMs in-place [1]. D-IMC leads to improvements on both energy efficiency and compute density compared to conventional solutions for various workloads [2-5]. Counter to analog IMC, D-IMC is immune to noise and non-idealities that reduce the precision of the analog MVM, enabling deterministic and repeatable MVMs. The D-IMC in Metis AIPU offers three advantages over previous approaches: (1) it stores multiple weight sets in the computational memory, enhancing storage density, allowing efficient accumulation of up to 16k inputs, and enabling simultaneous processing and weight reloading; (2) it maintains high energy efficiency at low utilization; and (3) it ensures full-precision accumulation, yielding competitive accuracy across workloads using post-training quantization without retraining.

The core component of the AI Core is the MVM engine, a 512×512 INT-8-8-26 D-IMC crossbar array. The array has a bit-serial activations feeder at the input and an integer arithmetic unit at the output, as shown in Fig. 11.3.4(a). The MVM engine is further segmented in 16 custom-designed IMC memory arrays (IMC banks), each of 512 input channels, 32 output channels, and 4 weight sets for a total memory capacity of 8 Mib. Each weight set, shown in Fig. 11.3.4(b), is independently addressed for write or compute operations. Simultaneous computation and weight reloading on different weight sets is also supported. The IMC bank operates bit-serially with each cycle processing 512 single bit activations through the input feeder from the L1 memory. Then, the results are accumulated over 8 cycles. External to the IMC bank, 26b accumulators are used to generate the final bit-accurate result. Finally, the integer arithmetic unit accumulates the partial products from large MVM operations without storing intermediate results back to memory. At each cycle, it returns a vector of 64 INT32 quantities, which the DPU utilizes for subsequent element-wise vector operations and activations. A crucial requirement of the MVM engine is to efficiently reduce the memory transactions of weights and inputs between the IMC banks and the memories. This is achieved by: (1) increasing the dimensions of the MVM engine, and (2) introducing multiple weight sets. For low-utilization workloads a clock and signal gating mechanism is implemented to improve energy efficiency, as illustrated in Fig. 11.3.4(c). The gating approach is divided in two parts: bank and block gating. At the MVM engine level each bank is clock-gated, thus the output gating granularity is a multiple of 32 channels. Within a bank, sets of 32 inputs are divided in 16 individually addressed clock-gated blocks. The inputs of the gated blocks are silenced to minimize the energy used in activation propagation buffers. Therefore, although the MVM engine supports a large matrix size, energy efficiency stays high even at low utilization.

A beta version of the Metis AIPU SoC is designed and fabricated in a 12 nm process to demonstrate the capabilities of the architecture. The functionality of the silicon implementation is shown in the plots in Figs. 11.3.5(a) and (b). The peak throughput is 57.3 TOPS at 0.7V and 875MHz. Nominally, a throughput of 54.2 TOPS is achieved at 800MHz. The energy efficiency for a variety of weight and activation distributions as a function of the supply voltage are depicted in Fig. 11.3.5(c). An energy efficiency of 15 TOPS/W at 0.68V is reported for random uniform activations and weights. Note that this efficiency is strongly dependent on the activation and the weight pattern. Under high sparsity conditions, e.g., 50% input and 90% weight sparsity, the energy efficiency reaches up to 82 TOPS/W. Figure 11.3.5(d) shows the impact of block gating on the efficiency. For all practical use cases and regardless of utilization, the energy efficiency remains high.

Table A in Fig. 11.3.5 lists performance measurements of the Metis AIPU SoC on several relevant neural networks like YoloV5. The results were obtained using an in-house experimental compiler, mapping execution of these networks on the multi-core AIPU architecture. The accuracy of the measured validation set closely aligns with the reference accuracy obtained using FP32 arithmetic. The number in parentheses in the accuracy column of Table A indicates the degree of similarity between the accuracy using FP32 and INT8. An energy efficiency of 354 FPS/W is achieved for ResNet50 running at 2502 FPS, while YoloV5s achieves 92 FPS/W at 497 FPS. At lower frequency, ResNet50 achieves an energy-efficiency of 512 FPS/W, delivering a throughput of 1430 FPS.

In Fig. 11.3.6, a comparison of our D-IMC with the state of the art is presented, detailing precision, throughput, compute density, and energy efficiency. Figure 11.3.7 shows a die photo and summarizes the key characteristics of the Metis AIPU silicon implementation.

Our Metis AIPU SoC is an integrated hardware solution optimized for AI inference, aiming computer vision workloads at the Edge. It integrates four homogeneous AI-Cores tailored for full neural network inference offloading. Each AI-Core is designed for self-reliance, allowing it to compute neural networks independently without external intervention. The performance of the AIPU is a result of a vast digital in-memory compute array, significantly larger than previous designs, coupled with gating techniques ensuring top-tier energy efficiency even in low utilization. Merging this expansive compute capability with an efficient memory subsystem and a versatile RISC-V control path, the Metis AIPU can simultaneously manage complex neural network tasks, setting a new standard for energy-efficient AI processing. This design achieves 15 to 82 TOPS/W of energy efficiency in the D-IMC and 353 FPS/W at 2502 FPS for ResNet-50 and 92 FPS/W at 497 FPS for YoloV5s at the SoC level.

References:

[1] A. Sebastian, et al., "Memory Devices and Applications for in-Memory Computing," *Nature Nanotechnology*, vol. 15, no. 7, pp. 529–544, 2020.
[2] D. Wang, et al., "DIMC: 2219TOPS/W 2569F2/b Digital In-Memory Computing Macro in 28nm Based on Approximate Arithmetic Hardware," *ISSCC*, pp. 266–267, Feb. 2022.
[3] H. Fujiwara, et al., "A 5-nm 254-TOPS/W 221-TOPS/mm² Fully-Digital Computing-in-Memory Macro Supporting Wide-Range Dynamic-Voltage-Frequency Scaling and Simultaneous MAC and Write Operations," *ISSCC*, pp. 186-187, Feb. 2022.
[4] F. Tu, et al., "ReDCIM: Reconfigurable Digital Computing-In-Memory Processor With Unified FP/INT Pipeline for Cloud AI Acceleration," *IEEE JSSC*, vol. 58, no. 1, pp. 243–255, Dec. 2022.
[5] C.-F. Lee, et al., "A 12nm 121-TOPS/W 41.6-TOPS/mm² All Digital Full Precision SRAM-based Compute-in-Memory with Configurable Bit-width For AI Edge Applications," *Symp. VLSI Technology and Circuits*, pp. 24–25, June 2022.

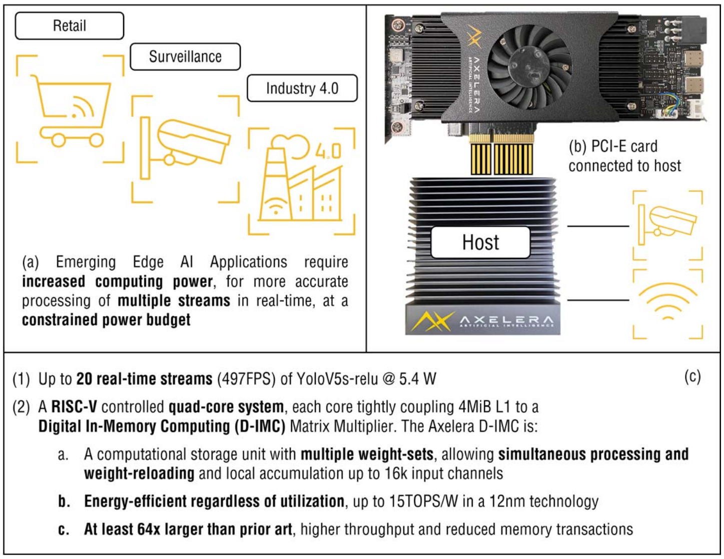

Figure 11.3.1: A solution for AI inference at the edge based on in-memory computing.

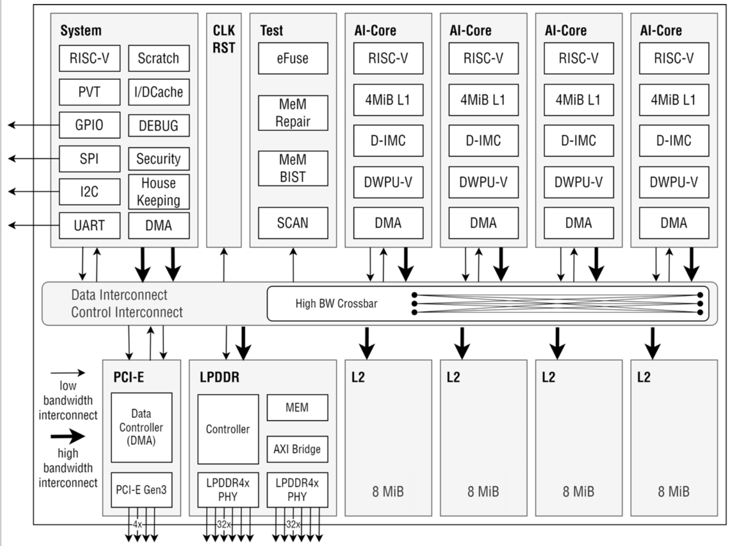

Figure 11.3.2: Metis AIPU SoC architecture.

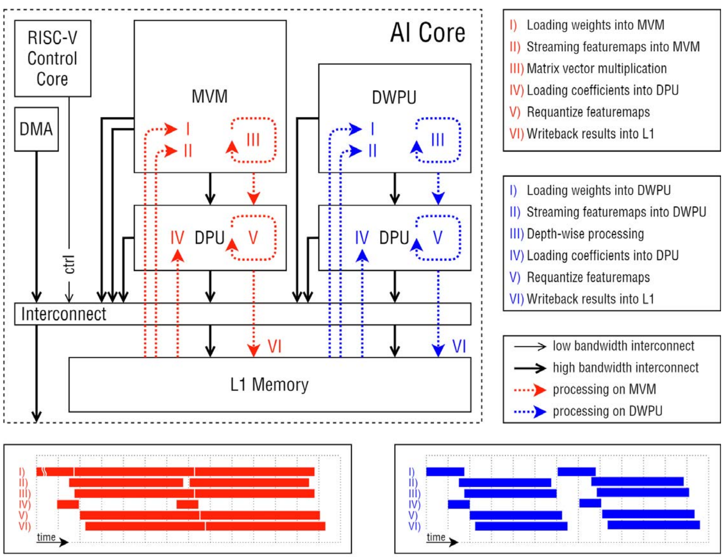

Figure 11.3.3: AI Core block diagram and operation mode.

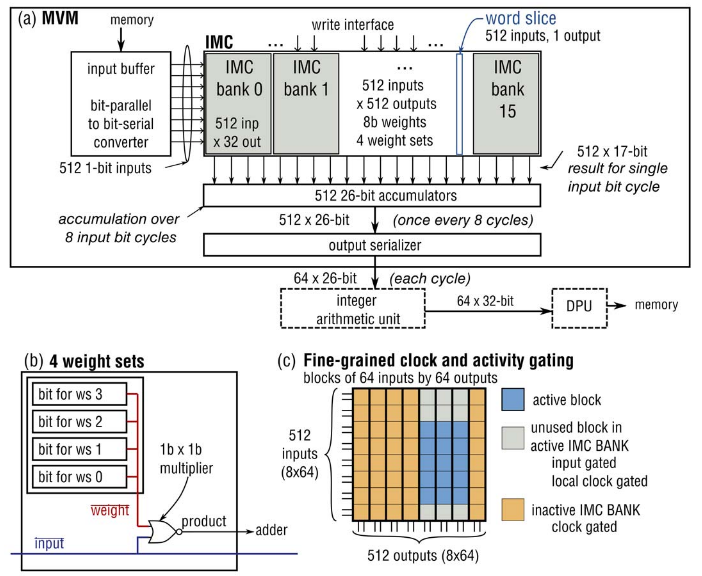

Figure 11.3.4: MVM engine block diagram, weight set implementation, and block gating.

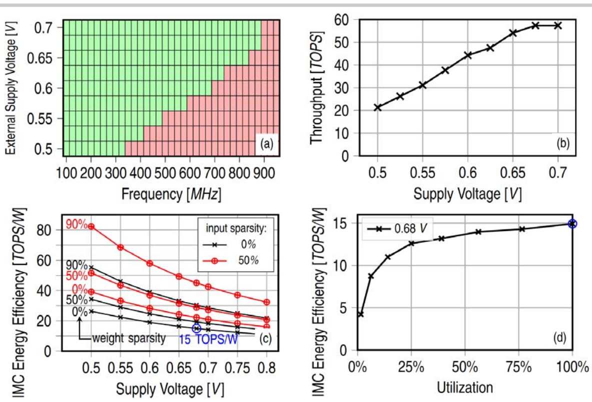

Table A: Metis Performance. Benchmarks run using experimental compiler.

Network	Resolution	Performance [FPS]	Accuracy@INT8	Chip Power [W]
ResNet-34	224×224	3199	73.2%* (-0.1)	7.1
ResNet-50	224×224	2502	76.0%* (-0.1)	7.1
SSD-MobileNetV1	300×300	5901	25.5 MAP+ (-0.3)	7.1
YoloV5s-ReLU [6]	640×640	497	33.3 MAP+ (-0.9)	5.4

* measured on ImageNet-1000 validation, + measured on COCO detection validation

[6] ultralitics, yolov5s-ReLU, 2021, *wandb.ai/glenn-jocher/activations/runs/22h4vude*

Figure 11.3.5: Measured performance of Metis D-IMC and Metis AIPU SoC.

Table B: Comparison to state-of-the-art

Metrics	Wang [2]	Fujiwara [3]	Tu [4]	Lee [5]	**Metis IMC**
	ISSCC 2022	ISSCC 2022	JSSC Jan. 2023	VLSI 2022	**This work**
Process Node	*28 nm*	*5 nm*	*28 nm*	*12 nm*	**12 nm**
IMC Area [mm^2]	0.033	0.0133	0.948	0.0323	**6.5**
Memory Size	*16 kb*	*64 kb*	*96 kb*	*8 kb*	**8 Mib**
Precision					
Weights (b_w) [*bits*]	1	4	8	4	**8**
Input Activations (b_{act}) [*bits*]	1	4	8	4	**8**
Output Activations [*bits*]	4	14	24	18	**26**
Performance	Normalized* for $8b \times 8b$ MACs**				
Power Supply [*V*]	0.45-1.1	0.9 / 0.5	0.6-1	0.72	**0.68**
Frequency [*MHz*]	250-280	1144 / 360	50-220	800	**800**
Throughput [*TOPS*]	0.3125	0.735 / 0.183	1.35	0.33	**52.4**
Compute Density [$TOPS/mm^2$]	9.46	55.3 / 13.8	1.43	10.39	**8**
Energy Efficiency [*TOPS/W*]	17.31	17.4*** / 63.3***	30.8	30.25	**15**

* Normalized performance is evaluated using: $\frac{\text{metric}}{(8/b_{act})(8/b_w)}$

** 1 *MAC* $= 2$ *OPS*

*** Using 10% input activity

Figure 11.3.6: Comparison of Metis D-IMC with the state of the art.

Table C: Core features of the Metis SoC.

Process	TSMC 12 *nm*
Die Area	12 × 12 mm^2
On-Chip L2 scratchpad	32 *MiB*
AI Cores	4
Support for LPDDR4x	
AI Core	
L1 scratchpad	4 *MiB*
IMC capacity	8 *Mib*
	512 x 512 x 4 weight sets
Weights	8-*bit* [signed and unsigned]
Activations	8-*bit* [signed and unsigned]
Outputs	32-*bit*
Throughput	52.4 *TOPS*
Power & Energy Efficiency	
Typical	5-15 *W*
IMC Energy Efficiency	15-82 *TOPS/W*
Off-Chip Connectivity	
High-speed interfaces	PCIe 3.0 4x
Low-speed interfaces	JTAG, SPI, I2C

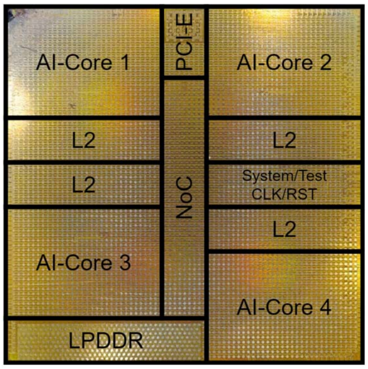

Figure 11.3.7: Core features of the Metis AIPU SoC and a die microphotograph.

